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250кит сдачи первого задания

1-3 недели II курса

прислать 0 и II

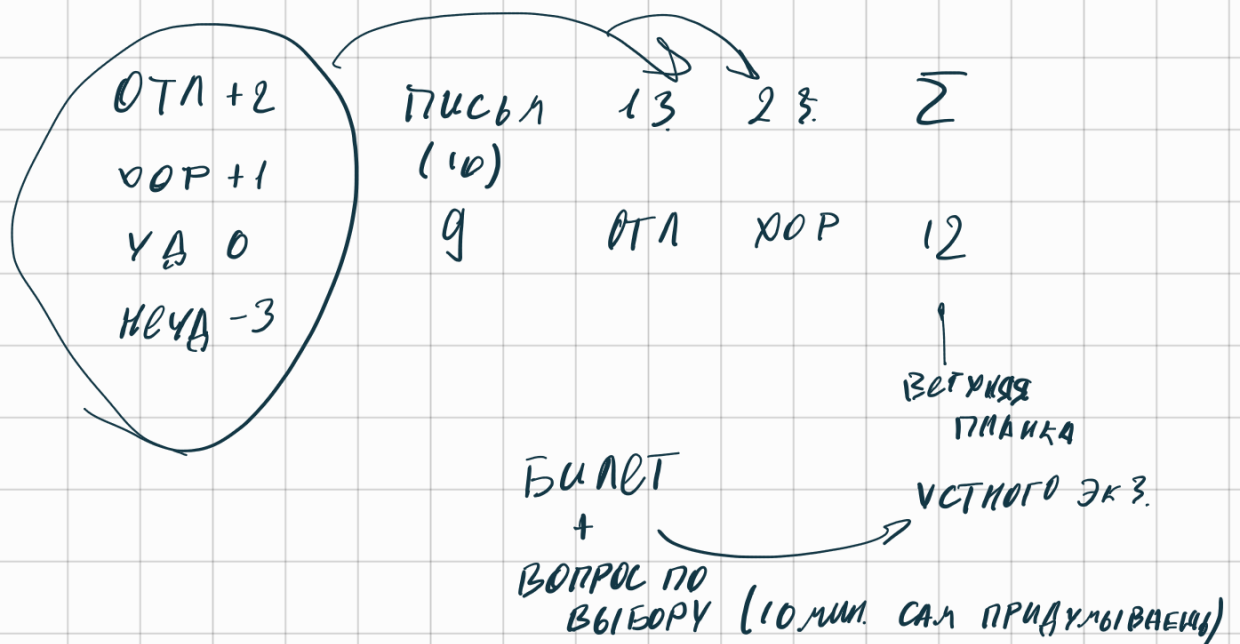
на к.р. печатать

Рисункок

Дано ; найти

ответ (формула + чис. отв.)

Физ. законы



302 ПК

Таблица производных	Таблица интегралов
1. $(C)' = 0$, $(x)' = 1$ $(x^2)' = 2x$	1. $\int dx = x + C$
2. $(x^a)' = ax^{a-1}$	2. $\int x^k dx = \frac{x^{k+1}}{k+1} + C$, $(k \neq -1)$
3. $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$, $\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$	3. $\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} + C$, $\int \frac{dx}{x} = \ln x + C$
4. $(a^x)' = a^x \ln a$, $(e^x)' = e^x$	4. $\int a^x dx = \frac{a^x}{\ln a} + C$, $\int e^x dx = e^x + C$
5. $(\log_a x)' = \frac{1}{x \ln a}$, $(\ln x)' = \frac{1}{x}$	5. $\int \frac{dx}{\sin x} = \ln \left \operatorname{tg} \frac{x}{2} \right + C$,
6. $(\sin x)' = \cos x$	6. $\int \frac{dx}{\cos x} = \ln \left \operatorname{tg} \left(\frac{x}{2} + \frac{\pi}{4} \right) \right + C$
7. $(\cos x)' = -\sin x$	7. $\int \sin x dx = -\cos x + C$
8. $(\operatorname{tg} x)' = \frac{1}{\cos^2 x}$	8. $\int \cos x dx = \sin x + C$
9. $(\operatorname{ctg} x)' = -\frac{1}{\sin^2 x}$	9. $\int \frac{dx}{\cos^2 x} = \operatorname{tg} x + C$
10. $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$	10. $\int \frac{dx}{\sin^2 x} = -\operatorname{ctg} x + C$
11. $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$	11. $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C$
12. $(\operatorname{arctg} x)' = \frac{1}{1+x^2}$	12. $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left x + \sqrt{x^2 \pm a^2} \right + C$
13. $(\operatorname{arcctg} x)' = -\frac{1}{1+x^2}$	13. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left \frac{x-a}{x+a} \right + C$
	14. $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \operatorname{arctg} \frac{x}{a} + C$

Правила дифференцирования и интегрирования		Некоторые тригонометрические формулы:	
а)	Правила дифференцирования	1	$\sin^2 x + \cos^2 x = 1$; $1 - \sin^2 x = \cos^2 x$; $1 - \cos^2 x = \sin^2 x$.
1	Производная суммы: $(u \pm v)' = u' \pm v'$	2	$\sin^2 x = \frac{1 - \cos 2x}{2}$, $\cos^2 x = \frac{1 + \cos 2x}{2}$,
2	Производная произведения: $(u \cdot v)' = u' \cdot v + u \cdot v'$	3	$\sin \alpha \cdot \sin \beta = \frac{1}{2} (\cos(\alpha - \beta) - \cos(\alpha + \beta))$

$$\vec{r}(t)$$

$$\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}$$

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \vec{r}'_t = \dot{\vec{r}} = \frac{d\vec{r}}{dt}$$

$$\vec{a} = \dot{\vec{v}} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \frac{d\vec{r}}{dt} = \frac{d^2\vec{r}}{dt^2} = \ddot{\vec{r}} \quad \text{или } \ddot{\vec{r}}$$

$$x(t) = x_0 + v_0 t + \frac{at^2}{2} \text{ полином. формул.}$$

$$a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$$

$$a dx = v dv$$

$$\int a dx = \int v dv$$

$$ax + C_1 = \frac{v^2}{2} + C_2$$

$$ax = \frac{v^2}{2} + C$$

$$ax_0 = \frac{v_0^2}{2} + C$$

$$C = ax_0 - \frac{v_0^2}{2}$$

$$ax = \frac{v^2}{2} + ax_0 - \frac{v_0^2}{2}$$

$$a(x - x_0) = \frac{v^2 - v_0^2}{2}$$

$$a dx = v dv$$

$$\int_{x_0}^x a dx = \int_{v_0}^v v dv$$

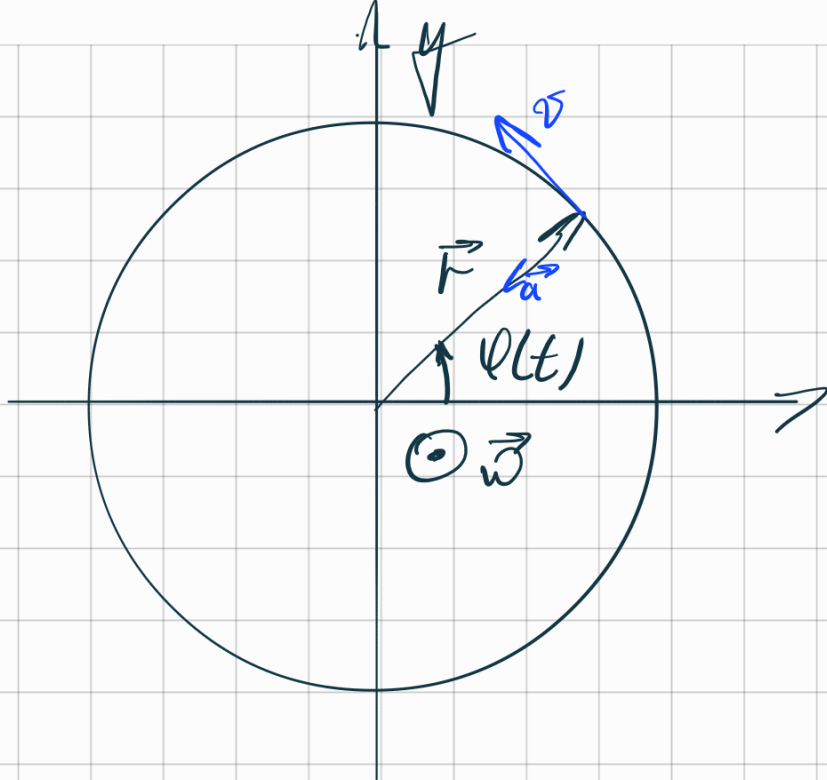
$$ax \Big|_{x_0}^x = \frac{v^2}{2} \Big|_{v_0}^v$$

$$a(x - x_0) = \frac{v^2 - v_0^2}{2}$$

$$x(t) = v_0 \cos \alpha t$$

$$y(t) = v_0 \sin \alpha t - \frac{gt^2}{2}$$

$$y = \operatorname{tg} \alpha \cdot x - \frac{g x^2}{2 v_0^2 \cos^2 \alpha}$$



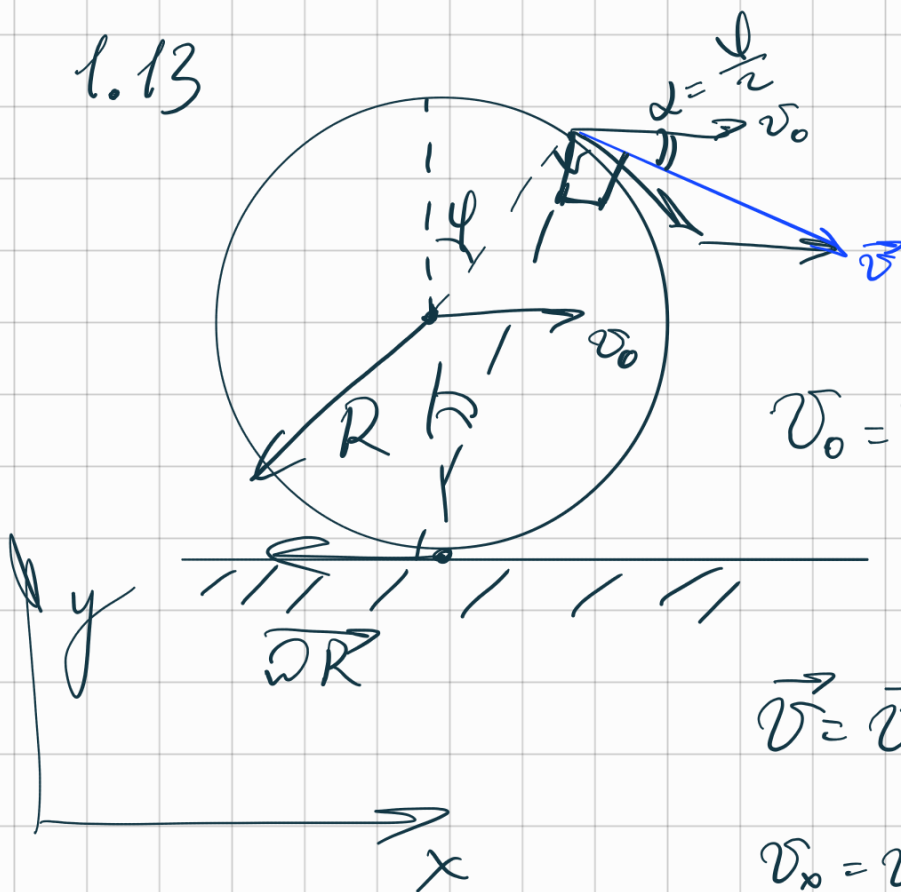
$$\dot{\omega} = \dot{\dot{\varphi}} - \text{yml. step.}$$

$$\epsilon = \dot{\omega} = \dot{\dot{\varphi}} - \text{yml. yml.}$$

$$\times \vec{v} = [\vec{\omega} \times \vec{r}]$$

$$\vec{a} = [\vec{\omega} \times \vec{v}]$$

1.13

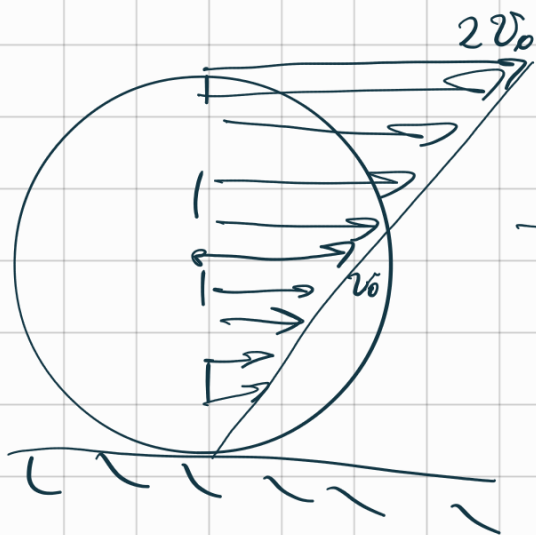


$$v_0 = \omega R \text{ - See previous.}$$

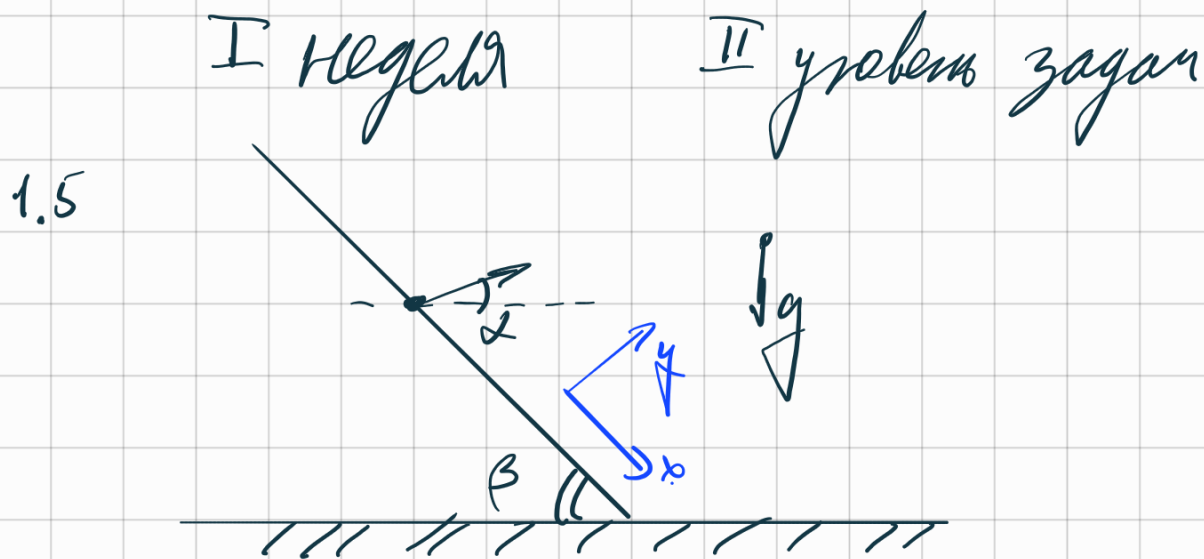
$$\vec{v} = \vec{v}_0 + [\vec{\omega} \times \vec{R}]$$

$$v_x = v_0 + v_0 \cos \psi$$

$$v_y = 0 + v_0 \sin \psi$$



$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{v_0^2(1 + \cos \psi)^2 + v_0^2 \sin^2 \psi} = v_0 \sqrt{2(1 + \cos \psi)}$$



Дано:

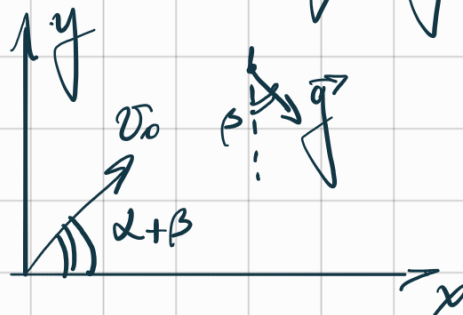
$$\beta = 45^\circ$$

Найти: α ; $x(\max)$

Решение:

введем С.О., где

x паралл. склону, а y перп. ему:



тогда в этих координатах

$$x = v_0 t \cos(\alpha + \beta) + \frac{g t^2}{2} \cdot \sin \beta$$

$$v_x = v_0 \cos(\alpha + \beta) + g t \cdot \sin \beta$$

$$y = v_0 t \sin(\alpha + \beta) - \frac{g t^2}{2} \cos \beta$$

$$v_y = v_0 \sin(\alpha + \beta) - g t \cdot \cos \beta$$

$$t = \frac{2 v_0 \sin(\alpha + \beta)}{g \cos \beta} \quad \text{— время полета}$$

$$x = v_0 \cos(\alpha + \beta) \cdot \frac{2 v_0 \sin(\alpha + \beta)}{g \cos \beta} + \frac{g \sin \beta}{2} \cdot \frac{4 v_0^2 \sin^2(\alpha + \beta)}{g^2 \cos^2 \beta}$$

$$\frac{x \cdot g \cdot \cos \beta}{v_0^2} = 2 \cos(\alpha + \beta) \cdot \sin(\alpha + \beta) + 2 \operatorname{tg} \beta \sin^2(\alpha + \beta) \quad (\beta = 45^\circ \Rightarrow \operatorname{tg} \beta = 1)$$

$$\frac{x g \cos \beta}{v_0^2} = \sin(2(\alpha + \beta)) + 2 \sin^2(\alpha + \beta) \quad \left| \cdot \frac{1}{2} \right. \leftarrow \text{узнаем новую}$$

$$0 = 2 \cos(2(\alpha + \beta)) + 4 \cos(\alpha + \beta) \sin(\alpha + \beta)$$

экстремума
функции $x(\alpha)$

$$0 = \cos(2(\alpha + \beta)) + \sin(2(\alpha + \beta)) \Leftrightarrow \operatorname{tg}(2(\alpha + \beta)) = -1$$

$$2(\alpha + \beta) = \frac{3\pi}{4}$$

или

$$2(\alpha + \beta) = -\frac{\pi}{4}$$

$$\alpha + \beta = \frac{3\pi}{8}$$

$$\beta = \frac{\pi}{4}$$

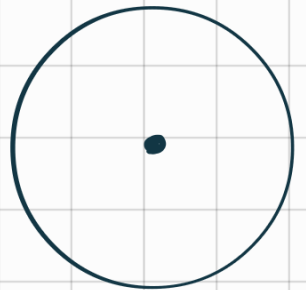
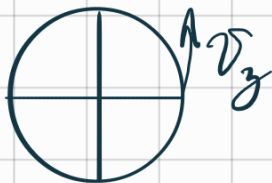
$$\alpha + \beta = -\frac{\pi}{8}$$

норм. корни
т.к. корни
бывают как
сложными

$$\alpha = \frac{\pi}{8}$$

$$\text{Ответ: } \frac{\pi}{8}$$

1.11



Дано:

$$R_n = 3,8 \cdot 10^5 \text{ км}$$

$$R_z = 6,38 \cdot 10^3 \text{ км}$$

$$T_z = 86164 \text{ с}$$

$$T_n = 2360591 \text{ с}$$

Найти: v_T

Решение:

Скорость тела v_T равна $v_T = v_n - v_z$,
где v_n - скорость тела по орбите, v_z - скорость
вращения поверхности Земли вокруг своей оси.

$$v_n = \frac{2\pi R_n}{T_n}$$

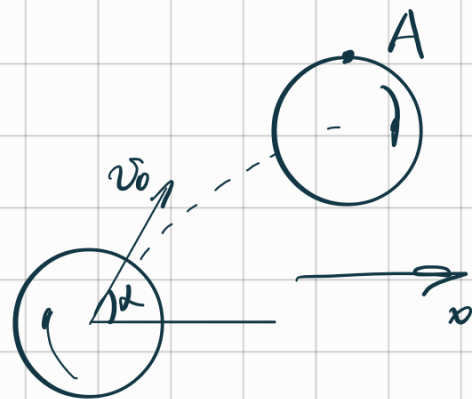
$$v_z = \frac{2\pi R_z}{T_z}$$

$$v_T = 2\pi \left(\frac{R_n}{T_n} - \frac{R_z}{T_z} \right) = 0,55 \frac{\text{км}}{\text{с}}$$

$$\text{Ответ: } 0,55 \frac{\text{км}}{\text{с}}$$

1.21

Решение:



Дано:

 $R; \omega; \alpha; v_0$ Найти: ρ

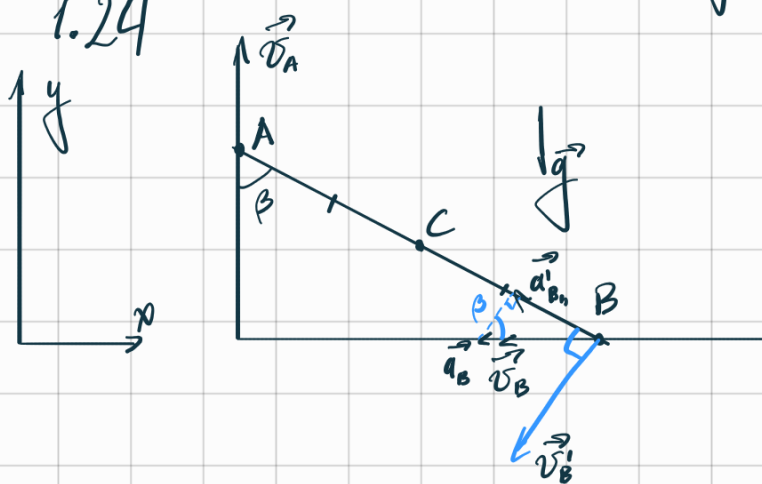
Т.к. диск находится в беззачетной точке, скорость его центра числ равна $v_0 \cos \alpha$, т.к. диск крутится по часовой стрелке и A находится в наивысшем положении (относительно ц.м.), то ее скорость относительно ц.м. (спроецированная на горизонт. ось Ox) равна ωR , а относительно земли $v_A = v_0 \cos \alpha + \omega R$ на A действует g и ц.с. ускорение $\omega^2 R$, тогда

$$a_A = g + \omega^2 R \left(\rho = \frac{v_A^2}{a_A} \right) \Rightarrow \rho = \frac{(v_0 \cos \alpha + \omega R)^2}{g + \omega^2 R}$$

(ц.с. ускорение сонаправлено с g)

Ответ: $\frac{(v_0 \cos \alpha + \omega R)^2}{g + \omega^2 R}$

1.24



Решение:

Т.к. $v_A = \text{const}$ введем ИСО, в которой A неподвижна

$$\vec{v}_B = \vec{v}_A + \vec{v}_B' \quad (\vec{v}_B' \perp l)$$

$$\vec{a}_B = \vec{a}_B'$$

спроецируем $\vec{v}_B$ на Oy :

$$v_A - v_B' \cdot \sin \beta = 0 \quad v_B' = \frac{v_A}{\sin \beta}$$

(стрелка вращается вокруг A)

$$a_{B_n}' = \frac{v_{B_n}'^2}{l} \quad a_{B_n}' = a_B' \cdot \sin \beta$$

$$a_B = a_B' = \frac{a_{B_n}'}{\sin \beta} = \frac{1}{\sin \beta} \cdot \frac{1}{l} \cdot \frac{v_A^2}{\sin^2 \beta} = \frac{v_A^2}{l \sin^3 \beta}$$

Дано:

 $l; v_A = \text{const}; \beta$ Найти: $\vec{a}_B$

Т.к. ступень "вращается" вокруг $\{A\}$, скорости и ускорения в любой его точке сонаправлены, тогда $a_C = \frac{1}{2} a_B = \frac{v_A^2}{2l \sin^3 \beta}$

$$\text{Ответ: } \frac{v_A^2}{2l \sin^3 \beta}$$

II неделя

0 уровень задач

2.1

Дано:

$$\dot{x} = \frac{x}{\tau}$$

$$\tau = 10 \text{ с.}$$

$$x_0 = 1 \text{ м.}$$

Найти: $x(t=\tau)$, $a(t=\tau)$

Решение:

$$1. \quad v = \frac{dx}{dt} = \frac{x}{\tau}$$

$$\frac{1}{x} dx = \frac{1}{\tau} dt$$

$$\ln(x) = \frac{t}{\tau} + C$$

$$\ln(x_0) = C$$

$$x_0 = 1 \Rightarrow C = 0$$

$$x = e^{\frac{t}{\tau}} \quad x(t=\tau) = e$$

$$2. \quad v = \frac{dx}{dt} = \frac{de^{\frac{t}{\tau}}}{dt}$$

$$v = \frac{1}{\tau} e^{\frac{t}{\tau}}$$

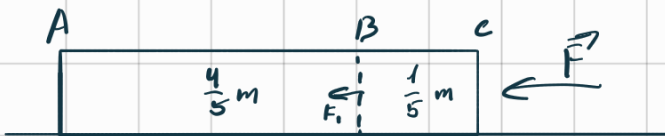
$$a = \frac{dv}{dt} = \frac{1}{\tau} \frac{de^{\frac{t}{\tau}}}{dt}$$

$$a = \frac{1}{\tau^2} \cdot e^{\frac{t}{\tau}}$$

$$a(t=\tau) = \frac{e}{\tau^2} = \frac{e}{100}$$

$$\text{Ответ: } e \text{ м.}; \frac{e}{100} \frac{\text{м}}{\text{с}^2}$$

2.1



Дано:

$$F; AB = \frac{4}{5} l$$

$$m; l$$

Найти: F_1

Решение:

$$a_{AC} = a_{BC} = a_{AB}$$

$$\vec{F}_{AB} = -\vec{F}_1 \quad F_{AB} = m_{AB} \cdot a_{AB}$$

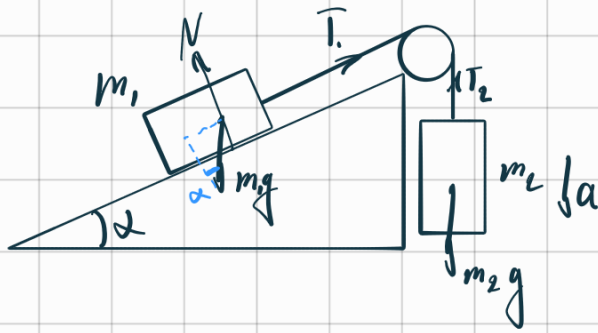
$$AB = \frac{4}{5} l \Rightarrow m_{AB} = \frac{4}{5} m$$

$$F_{AB} = \frac{4}{5} m a_{AC} = \frac{4}{5} F$$

$$F_1 = -\frac{4}{5} F$$

$$\text{Ответ: } -\frac{4}{5} F$$

2.5



Дано:

 $m_1; m_2; \alpha$ Искомое: T, a (считаем трение нет,
т.е. идеальное тело)

Решение:

$$T_1 = T_2 = T$$

Из уравнений, что m_2
опускается:

$$m_2 a = m_2 g - T$$

$$m_1 a = T - m_1 g \cdot \sin \alpha$$

$$a(m_2 + m_1) = g(m_2 - m_1 \sin \alpha)$$

$$a = g \frac{m_2 - m_1 \sin \alpha}{m_1 + m_2}$$

$$T = m_2 g - m_2 g \frac{m_2 - m_1 \sin \alpha}{m_1 + m_2} = m_2 g \left(\frac{m_1 + m_2 - m_2 + m_1 \sin \alpha}{m_1 + m_2} \right) = \frac{m_1 m_2 g}{m_1 + m_2} (1 + \sin \alpha)$$

$$\text{Ответ: } g \frac{m_2 - m_1 \sin \alpha}{m_1 + m_2}; \frac{m_1 m_2 g}{m_1 + m_2} (1 + \sin \alpha)$$

$$1.17 \quad \int_0^T |v| dt = \int v_0 \sqrt{2(1+\cos\varphi)} dt =$$

$$= v_0 \sqrt{2 \frac{1+\cos\varphi}{2}} dt =$$

$$\varphi(t) = \omega t - \pi$$

$$d\varphi = \omega dt$$

$$= 2v_0 \int_0^T \cos \frac{\varphi}{2} dt =$$

$$\cos \frac{\varphi}{2} = \sqrt{\frac{1+\cos\varphi}{2}}$$

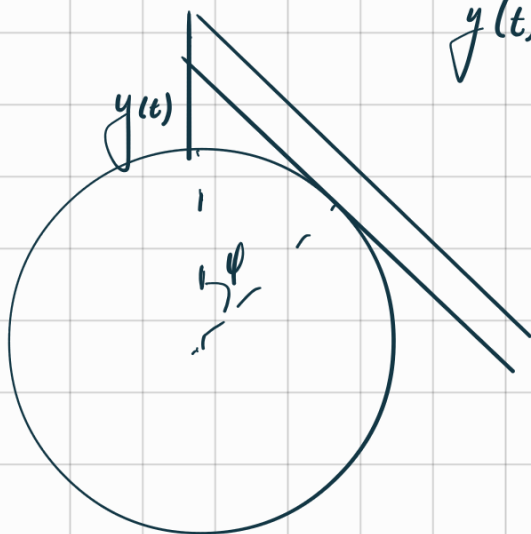
$$= 2v_0 \int_0^T \cos \frac{\omega t - \pi}{2} dt =$$

$$= 2v_0 \sin \frac{\omega t - \pi}{2} \cdot \frac{2}{\omega} \Big|_0^T =$$

$$= 4 \frac{v_0}{\omega} \left(\sin \frac{\omega T - \pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right) =$$

$$= 4R \left(\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right) = 8R$$

1.12



$y(t) = ?$

$$\dot{\varphi} = \frac{2\pi}{T_y}$$

$$\cos\varphi = \frac{R}{R+y}$$

$$y = R \frac{1 - \cos\omega t}{\cos\varphi \omega t}$$

$$y \approx R(1 - \cos\omega t)$$

$$\dot{y} = R \sin\omega t \cdot \omega \quad \ddot{y} = \omega^2 R \cos\omega t = 3,4 \frac{\text{cm}}{\text{s}^2}$$

$$H = \frac{\dot{y} \tau}{2}$$

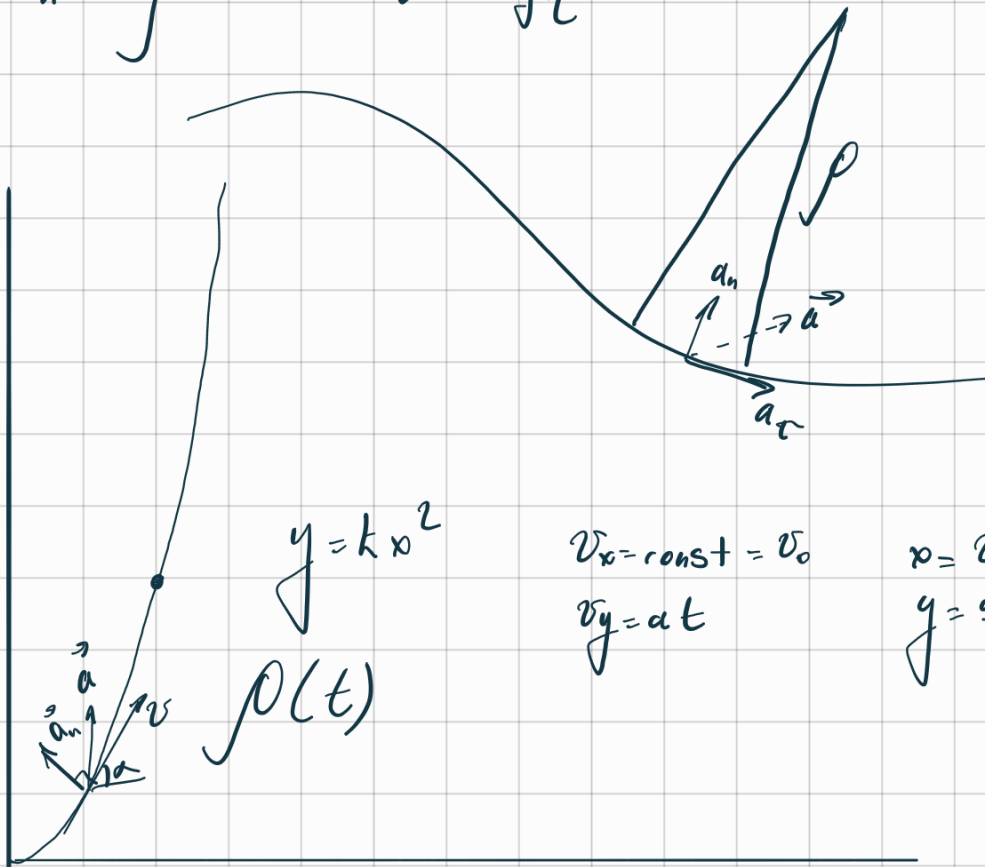
$$T = \sqrt{\frac{2H}{\omega^2 R}} \approx 5 \text{ mm}$$

$$\cos \omega t = 1 - \frac{(\omega t)^2}{2}$$

$$y = R \frac{\frac{(\omega t)^2}{2}}{1 - \frac{(\omega t)^2}{2}} = R \frac{(\omega t)^2}{2} \cdot \frac{2}{2 - (\omega t)^2} = R \frac{(\omega t)^2}{2 - (\omega t)^2}$$

$$a_n = \frac{v^2}{\rho}$$

$$a_\tau = \frac{dv}{dt}$$



$$y = kx^2$$

$$\rho(t)$$

$$v_x = \text{const} = v_0$$

$$v_y = at$$

$$x = v_0 t$$

$$y = \frac{at^2}{2} = kx^2 = v_0^2 t^2 k$$

$$v_0 = \sqrt{\frac{a}{2k}}$$

$$v^2 = v_x^2 + v_y^2 = \frac{a}{2k} + a^2 t^2$$

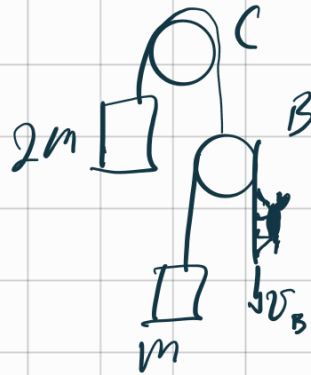
$$a_n = a \cos \alpha$$

$$\cos \alpha = \frac{v_x}{v}$$

$$\vec{p} = m \vec{v}$$

$$m \vec{a} = m \frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt} = \vec{F}$$

2.17



Условия:

$$T_1 - mg = ma_1$$

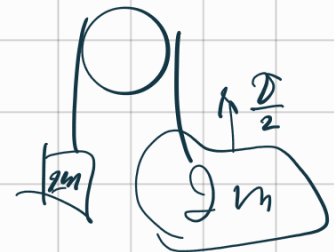
$$T_2 - mg = ma_2$$

$$T_1 = T_2 = T$$

(нить нерастяжима, невесомая)

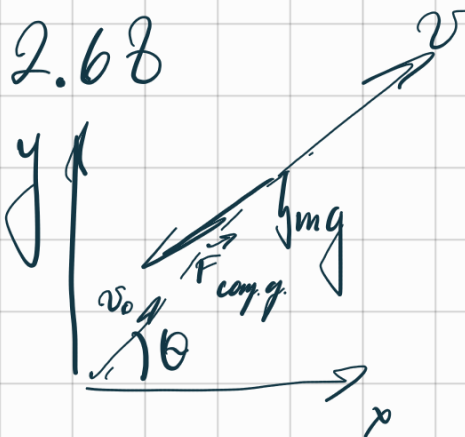
$$v_{\text{отн B}} = \frac{v_B}{2}$$

$$v_B - v_{\text{отн B}} = v_{\text{отн B}}$$



$$v_{2m} = \frac{v}{4}$$

2.68



$$t(K_{\max})$$

$$L(K_{\max})$$

$$m \frac{dv_x}{dt} = -\beta v_x$$

$$m \frac{dv_y}{dt} = -\beta v_y - mg$$

$$v_x(t) = v_0 \cos \theta e^{-\frac{\beta t}{m}}$$

$$x(t) = \int_0^t v_0 \cos \theta e^{-\frac{\beta t}{m}} dt =$$

$$= \frac{m v_0}{\beta} \cos \theta (1 - e^{-\frac{\beta t}{m}})$$

$$m \frac{dv_y}{dt} = -\beta (v_y + \frac{mg}{\beta})$$

$$\frac{dv_y + \frac{mg}{\beta}}{v_y + \frac{mg}{\beta}} = -\frac{\beta}{m} dt$$

$$\ln(v_y + \frac{mg}{\beta}) = -\frac{\beta}{m} t$$

$$\ln v_y + \frac{mg}{\beta}$$

2.43



$$F_{TP} = \mu N$$

$$\Pi_3.11$$

$$ma_n = N = m \frac{v^2}{R}$$

$$ma_\tau = -F_{TP} = -\mu \cdot N = -\mu m \frac{v^2}{R}$$

$$\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx} = -\mu \frac{v^2}{R}$$

$$\int \frac{dv}{v} = -\int \frac{\mu}{R} dx$$

$$\ln \frac{v}{v_0} = -\frac{\mu l}{R}$$

$$v = v_0 e^{-\frac{\mu l}{R}}$$

$$v_k = v_0 e^{-\pi \mu}$$

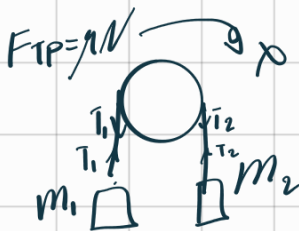
$$\frac{dv}{dt} = -\mu \frac{v^2}{R}$$

$$\int_{v_0}^v \frac{dv}{v^2} = \int \frac{-\mu}{R} dt$$

$$\frac{1}{v_0} - \frac{1}{v} = -\frac{\mu t}{R}$$

$$v(t) = \frac{v_0 R}{R + v_0 \mu t}$$

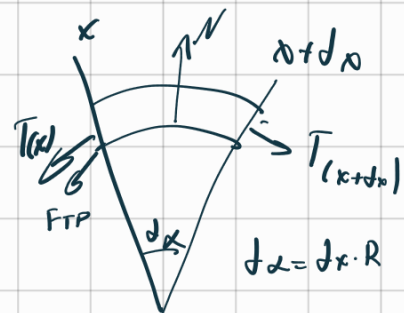
2.57



б. габарити

$$T_1 = m_1 g$$

$$T_2 = m_2 g$$



$$dx = dx \cdot R$$

КАСАТЕЛЬНАЯ

$$T(x+dx) \cos \frac{dx}{2} = T(x) \cos \frac{dx}{2} + F_{TP}$$

$$dT = T(x+dx) - T(x) = \mu N$$

НОРМАЛЬ

$$N = T_x \sin \frac{dx}{2} + T_{(x+dx)} \sin \frac{dx}{2} = T dx$$

$$dT = \mu T dx$$

$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^{\pi} \mu dx$$



$$\frac{T_2}{T_1} = e^{\mu\pi} = \frac{m_2(g-a)}{m_1(g+a)}$$

в равновесии

$$e^{-\pi\mu} \leq \frac{m_2}{m_1} \leq e^{\pi\mu}$$

$$\vec{F} = \frac{d\vec{P}}{dt} = \frac{d(m\vec{v})}{dt}$$

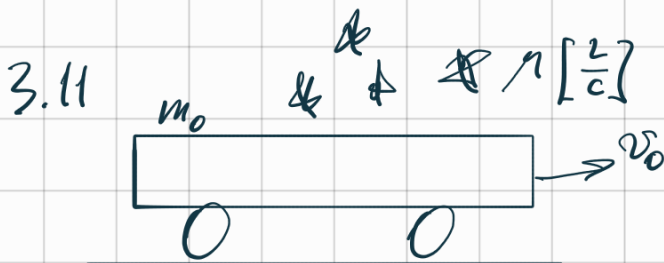
$$m \frac{d\vec{v}}{dt} = \vec{F} + \vec{u} \frac{dm}{dt}$$

у-е реактивного



$$m = m_0 e^{-\frac{v}{u}} - \text{ф-ла Циолковского}$$

$$v = u \ln \frac{m_0}{m}$$



1) через нарастаем

$$m(t) = m_0 + \mu t$$

$$m_0 v_0 = (m_0 + \mu t) u$$

$$u = \frac{m_0 v_0}{m_0 + \mu t} \xrightarrow{t \rightarrow \infty} 0$$

$$x(t) = \frac{m_0 v_0}{\mu} \int \frac{dt}{\frac{m_0}{\mu} + t} = \frac{m_0 v_0}{\mu} \ln \left(1 + \frac{\mu t}{m_0} \right) \xrightarrow{t \rightarrow \infty} \frac{m_0 v_0}{\mu} \ln \left(\frac{\mu t}{m_0} \right)$$

2) через сохраняем

$$dP = F dt = 0$$

$$(m_0 + \mu t)(v + dv) = m_0 v$$

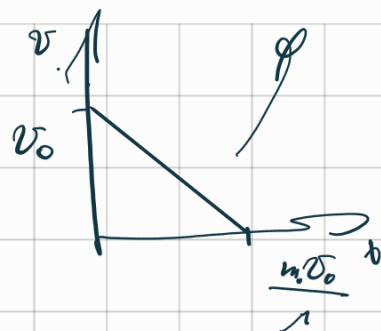
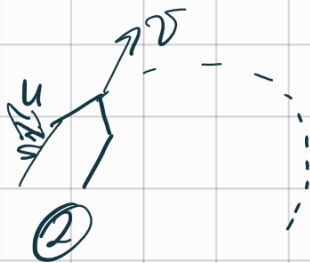
$$m_0 v + m_0 dv + \mu v dt + \mu t dv = m_0 v$$

$$m_0 dv = -\mu v dt = -\mu dx \quad m_0 v = -\mu x$$

$$\frac{dv}{v} = -\frac{\mu}{m_0} dt$$

$$v_{\infty} = v_0 e^{-\frac{\mu t}{m_0}} \xrightarrow{t \rightarrow \infty} 0$$

3.43



$$1) \underbrace{(m + \delta m)(v + \delta v)}_{P_{\text{ПРК}}} + \underbrace{(-\delta m)(v + u)}_{P_{\text{ТОЛЧБА}}} - m v = 0$$

$$m \delta v - \delta m u = 0$$

$$\ln \frac{m}{m_0} = \frac{1}{u} (v - v_0)$$

$$m_1 = m_0 e^{-\frac{2v_0}{u}}$$

$$2) (m + \delta m)(\vec{v} + \delta \vec{v}) + (-\delta m)(\vec{v} + \vec{u}) = m \vec{v}$$

$$m \delta \vec{v} - \delta m \vec{u} = 0$$



$$-m \delta v - u \delta m = 0$$

$$\delta x = \frac{\delta v}{v_0}$$

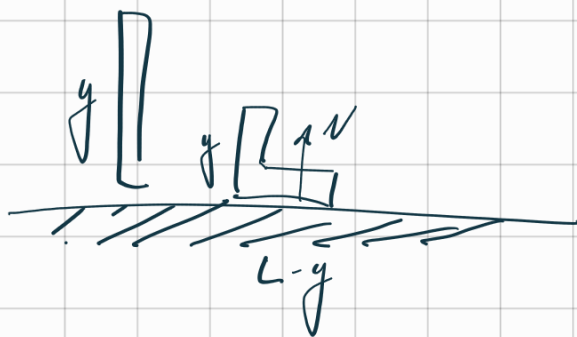
$$\bullet -m v_0 \delta x = u \delta m$$

$$\int_{m_0}^m \frac{\delta m}{m} = - \int_{v_0}^v \frac{\delta v}{u}$$

$$\ln \frac{m}{m_0} = -\frac{u v_0}{u^2}$$

$$m = m_0 e^{-\frac{v v_0}{u}}$$

4.55



$$v = m \frac{L-y}{L} g + \frac{d P_{y1}}{dt} = m \frac{L-y}{L} g + \frac{dm \cdot v}{dt} = m \frac{L-y}{L} g + \frac{1}{L} \frac{dy}{dt} \cdot \frac{2g(L-y)}{v_y} = m \frac{L-y}{L} g + \frac{m}{L} \cdot 2g(L-y)$$

$$= 3 \frac{m}{L} (L-y) g$$

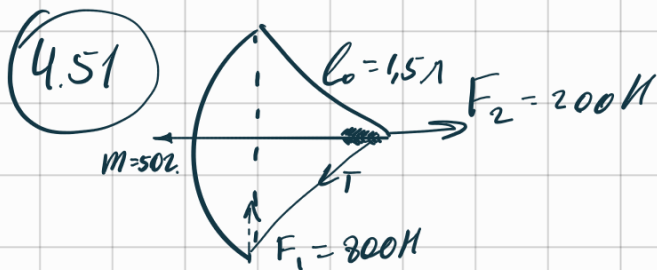
$$\delta A = (\vec{F} \cdot d\vec{r})$$

$$A = \int \vec{F} \cdot d\vec{r}$$

$$\delta a = m \frac{d\vec{v}}{dt} \cdot d\vec{r} = m \vec{v} \cdot d\vec{v} = \int \frac{m v^2}{2} = \int \left(\frac{p^2}{2m} \right) = \int K$$

$$\Delta K = A_{\text{внешн}}$$

Т.е. из А кинетическая энергия увеличивается



$$F = \text{const} = T = F_1$$

$$F(x) = 2T \sin \varphi = 4 F_1 \frac{x}{L_0} \quad x_{\text{max}} = \frac{F_2 L_0}{4 F_1}$$

$$A = \int_0^{x_{\text{max}}} F_x dx = \frac{F_2^2 L_0}{8 F_1} = \frac{m v_0^2}{2}$$

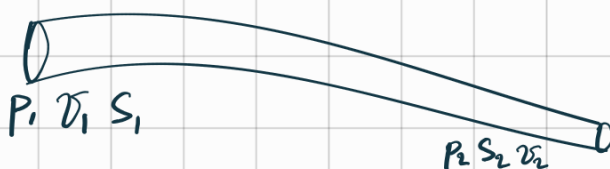
$$L_{\text{max}} = \frac{v_0^2}{g} = \frac{F_2^2 L_0}{4 F_1 m g} = \frac{100^2}{4 \cdot 200 \cdot 0,05 \cdot 10} = \frac{100 \cdot 45}{40} = 25 \cdot 4,5 = 37,5$$

$$\Delta K = A_{\text{внешн}} = A_{\text{конк}} + A_{\text{внешн}} = -\Delta U + A_{\text{внешн}}$$

$$\exists \text{ концеп. F} \quad \delta U = -\delta A$$

$$\Delta E = \Delta(K+U) = A_{\text{внешн}}$$

И СНОВА ГИДРОДИНАМИКА...



Умножим на m

$$\frac{dm_1}{dt} = - \frac{dm_2}{dt}$$

$$\frac{m}{2} (v_2^2 - v_1^2) + (u_2 - u_1) = P_1 S_1 dx_1 - P_2 S_2 dx_2$$

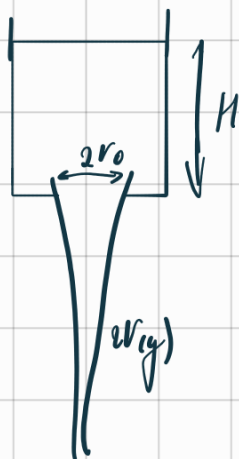
$$\frac{v_2^2}{2} - \frac{v_1^2}{2} + \frac{u_2 - u_1}{m} = \frac{P_1}{\rho_1} - \frac{P_2}{\rho_2}$$

$$\frac{P_1}{\rho_1} + \frac{u_1}{m} + \frac{v_1^2}{2} = \frac{P_2}{\rho_2} + \frac{u_2}{m} + \frac{v_2^2}{2} = \text{const}$$

$$\frac{\rho_1 S_1 dx_1}{dt} = \frac{\rho_2 S_2 dx_2}{dt}$$

$$\rho_1 S_1 v_1 = \rho_2 S_2 v_2$$

14.11



$$P_0 - 0 + \rho \frac{v^2}{2} = P_0 - \rho g y + \frac{\rho v_y^2}{2} = P_0 + \rho g H$$

$$\pi r_0^2 v_0 = \pi r_y^2 v_y$$

$$v_y = \sqrt{2g(H+y)}$$

$$v_y = r_0 \sqrt{\frac{v_0}{v_y}} = r_0 \sqrt{\frac{H}{H+y}}$$

BCL 🤪

Угловы мом

$$\vec{r}_c = \frac{\sum m_i \vec{r}_i}{\sum m_i} \rightarrow \frac{\sum \vec{r} dm}{M}$$

$$\vec{v}_c = \frac{\sum m_i \vec{v}_i}{\sum m_i} = \frac{\vec{P}}{M}$$

$K_{\text{обм}} = K_c + K_{\text{отн}}$ - Теорема Кёнига

1) $K_{\text{отн}} \rightarrow 0$ для неупр. удара

2) $K_{\text{отн}} = \text{const}$ для упр. удара

4.25 $\begin{bmatrix} \frac{3}{4} \\ \frac{1}{4} \end{bmatrix} \begin{matrix} \text{--- } y_{12} \\ \text{--- } y_{11} \end{matrix}$

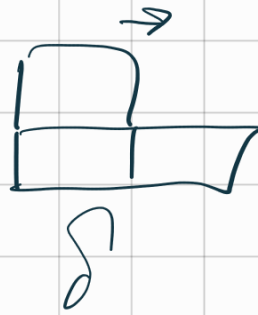
$$t = \sqrt{\frac{2L}{g}} = \sqrt{\frac{L}{g}}$$

$t' = \sqrt{2} t$ - время пути
поверхности

4.47



a)



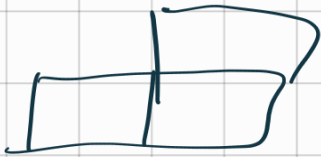
$$P_a = P_b$$

$$m v_0 + m v_0 = 2m v + m v_0$$

$$v = \frac{v_0}{2}$$

$$A_{TP} = \int_0^l F_{TP} dx = \int_0^l k x dx = \int_0^l k mg \frac{x}{2} dx = k mg \frac{l}{2}$$

2)



$$P_2 = P_1$$

$$2m v + m v_0 = 3m v_k$$

$$v_k = \frac{2}{3} v_0$$

$$A_{TP} = -\Delta K_{TD} = -\left(\frac{3}{2} m \left(\frac{2}{3} v_0 \right)^2 - \frac{2}{2} m \left(\frac{v_0^2}{2} \right) - m \frac{v_0^2}{2} \right) = \frac{m v_0^2}{12}$$

$$l = \frac{v^2}{6kg}$$